fighters such as doctors, in case of a confirmed case.

ETHEREAL MACHINES

Indian healthcare systems are notoriously under-equipped when it comes to having an adequate amount of medical supplies for health emergencies. Especially one at the gargantuan scale of the extremely infectious COVID-19. There are only about 40,000 ventilators available across India and if the positive cases of the novel coronavirus touch anywhere close to the numbers in the US and Italy, the outcome could be catastrophic. To combat the lack of ventilators, several startups have begun manufacturing ventilators, however, Ethereal Machines, a Bengaluru-based startup, has come up with a unique solution to meet the demand. Founders Kaushik Mudda and Navin Jain developed a 3D printed ventilator splitter which can ventilate two patients simultaneously even though they have different ventilatory needs. This is accomplished via differential pressure splitting. A single ventilator can supply two patients with different supplies of oxygen via the splitter which divides the oxygen as needed. This cuts the number of ventilators needed in half and could be path-breaking during this pandemic. Mudda and Jain worked closely with a team of doctors at Aster Hospitals' Sonal Asthana. The team successfully created a device that could achieve splitter ventilation and also prevent cross-contamination between paired patients. Ethereal Machines then improved upon the design further. The company believes that its design increases the efficacy of a ventilator

Ethereal Machines are 3D printing ventilator splitters to meet the increasing demand

while decreasing the risks of simple splitters. This company won a ₹50 lakh grant from the Action Covid-19 Team (ACT).

KARKHANA.IO

This is yet another Indian startup employing 3D printing to mass-manufacture PPEs (personal protective equipment). PPEs are the first line of defence for healthcare professionals that are toiling tirelessly to treat COVID-19 patients. As of now, the country has a concerning shortage of PPEs. Therefore, to meet the demand, Mumbai-based startup, Karkhana.io is using technologies such as 3D printing, injection moulding, machining, and fabrication to rapidly increase the production of PPEs. The startup has begun manufacturing face shields, goggles and aerosol boxes. As of now, Karkhana.io is producing 1200 face shields per day but they plan on ramping up production significantly where they are producing 5000 shields per day. According to the company, the face shields can be reused if disinfected properly. The company has already started receiving orders from the government and private hospitals. It is also supplying other manufacturers with tools to mass-produce PPEs and ICU equipment such as connectors, valves and ventilator parts. Karkhana.io also received a grant from the ACT of ₹20 lakh.

VEDANTU

Edtech startup, Vedantu, is coming to the aid of students by providing online learning for free on their website. The startup is allowing students to access their entire suite of learning courses for free in an effort to allow students to continue learning while staying safe within the confines of their homes. The website caters to students from Grade 1 to Grade 12 by providing classes in subjects such as Math, English, EVS, Science and SST and PCMB (physics, chemistry, maths, biology) for classes 11 and 12 as well as JEE and NEET. Students also gain free access to a host of co-curricular courses for all grades such as Turbo Maths, Rocket Pro, Photography, Coding, Grammar and more. The platform even features live classes and all their content can be accessed on mobile phones, PCs or tablets. Additionally, Vedantu is even extending support to several schools to teach their students remotely. The school teachers can reach out to the company to use its Wave platform to teach students online.

PERSAPIEN

Founded by Stanford University researchers, Debayan Saha and Shashi Ranjan, Delhi-based startup, PerSapien has developed 'Airlens Minus Corona', a huge human-shaped robot that is capable of sterilising spaces in cities such as streets, bus stands, shopping malls, hospitals,

Airlens Minus Corona can disinfect entire cities

railway stations, and more to restrict the spread of COVID-19. The Airlens Minus Corona electrifies or ionises water droplets to kill COVID-19 viral proteins. The ionisation process oxidises the viral proteins into harmless molecules. Saha and Ranjan believe that their oxidation technique can potentially sterilise an entire city. The device is based on the technology used in PerSapien's Minus-2point5, a device that combats air pollution. With the outbreak of the COVID-19 pandemic, the duo decided to repurpose their existing technology to fight against the virus. Saha and Ranjan have already reached out to the government concerning the implementation of their technology in COVID-19 hotspots. The duo also decided not to sell the devices as they intend to make the technology available for the service of citizens in a cost-effective manner. d